

Day 04 - Introduction to Linux

Objective

To understand the fundamentals of Linux, its importance in Cloud Computing and DevOps, explore popular Linux distributions, install Linux using WSL, and practice essential Linux terminal commands.

What is Linux?

Linux is a free and open-source operating system based on the Linux kernel, created by Linus Torvalds in 1991. It is widely used in servers, cloud platforms, embedded systems, and software development due to its stability, security, and flexibility.

Key Features

- Open Source
 - Secure
 - Stable
 - Multi-user
 - Multitasking
 - Highly customizable
 - Command-line support
 - Efficient resource management
-

Why Linux is Widely Used in the IT Industry

Linux powers most of the world's servers and cloud infrastructure.

Reasons

- High performance
- Excellent security
- Low resource usage
- Reliable for 24/7 systems
- Preferred by cloud providers
- Strong community support
- Free to use

Common Uses

- Web Servers
 - Database Servers
 - Cloud Computing
 - DevOps
 - Cybersecurity
 - Containers (Docker)
 - Kubernetes Clusters
-

Advantages of Linux

Linux	Other Operating Systems
Open Source	Mostly Proprietary
Free	Usually Paid
Highly Secure	More Vulnerable to Malware
Lightweight	Higher Resource Usage
Ideal for Servers	Mostly Desktop Focused
Powerful Command Line	GUI Focused

Linux Distributions

A Linux Distribution (Distro) is an operating system built using the Linux kernel along with software packages and tools.

Popular Linux Distributions

Ubuntu

- Beginner friendly
- Most commonly used in Cloud Computing
- Excellent community support

Debian

- Stable and secure
- Preferred for production servers

CentOS

- Enterprise Linux distribution
- Traditionally used in enterprise environments

Fedora

- Latest Linux technologies
- Community-driven

Arch Linux

- Lightweight
- Highly customizable
- Advanced users

Installing Linux Using WSL

What is WSL?

WSL (Windows Subsystem for Linux) allows Linux to run directly on Windows without requiring a virtual machine or dual boot.

Benefits

- Easy installation
- Native Linux terminal
- Low resource usage
- Perfect for development
- Supports Linux commands on Windows

Installation Steps

1. Enable WSL.
2. Install Ubuntu from the Microsoft Store.
3. Restart the computer if required.
4. Create a Linux username and password.
5. Open the Ubuntu terminal and start using Linux commands.

Basic Linux Terminal Commands

1. Print Working Directory

```
pwd
```

Displays the current directory.

2. List Files and Folders

```
ls
```

Shows files and directories in the current location.

3. Change Directory

```
cd folder-name
```

Moves to another directory.

4. Create a Directory

```
mkdir folder-name
```

Creates a new directory.

5. Remove a Directory

```
rmdir folder-name
```

Deletes an empty directory.

6. Create a File

```
touch filename.txt
```

Creates an empty file.

7. Copy Files

```
cp source destination
```

Copies files or folders.

8. Move or Rename Files

```
mv oldname newname
```

Moves or renames files.

9. Delete Files

```
rm filename
```

Deletes files.

10. Clear Terminal

```
clear
```

Clears the terminal screen.

11. Display File Contents

```
cat filename
```

Displays the contents of a file.

12. Current User

```
whoami
```

Shows the currently logged-in user.

13. System Information

```
uname -a
```

Displays detailed information about the Linux system and kernel.

Commands Practiced Today

```
pwd  
ls  
cd  
mkdir  
rmdir  
touch  
cp  
mv  
rm  
clear  
cat  
whoami  
uname -a
```

Hands-On Practice

- Installed Ubuntu using WSL
- Opened the Linux terminal
- Navigated directories
- Created folders
- Created files
- Copied files
- Renamed files
- Deleted files

- Displayed file contents
 - Checked current user
 - Viewed system information
-

Key Takeaway

Linux is more than an operating system—it is the backbone of modern Cloud Computing and DevOps. Most cloud servers are managed using the command line, making Linux terminal skills essential for deploying, managing, and troubleshooting applications in production environments.

Day 04 Summary

Today I learned the fundamentals of Linux, explored different Linux distributions, installed Ubuntu using WSL, and practiced essential Linux commands. These basics provide the foundation for learning shell scripting, server administration, Docker, Kubernetes, and cloud platforms like AWS in the coming days.

Status:  Day 04 Completed